

**Course Plan**

Semester: 7

Year:2026-27

|                                                                       |                        |
|-----------------------------------------------------------------------|------------------------|
| Course Title: Big Data Analytics                                      | Course Code: 26ECSC404 |
| Total Contact Hours: 58 Hrs                                           | Duration of ESA: 2hrs  |
| ISA Marks: 50                                                         | ESA Marks: 50          |
| Lesson Plan Author: Mr.Channabasappa Muttal, Dr.Krupashankari Sandyal | Date: 22/7/2026        |
| Checked By: Dr.Suvarna G Kanakaraddi                                  | Date: 22/7/2026        |

**Brief description of the course:**

This course provides an in-depth understanding of terminologies and the core concepts behind big data problems, applications, systems and the techniques, that underlie today's big data computing and storage technologies. It provides an exposure on some of the most common frameworks such as Apache Spark, Hadoop, MapReduce, Large scale data storage technologies such as in-memory key/value storage systems, NoSQL distributed databases, Big Data Streaming Platforms such as Apache Spark Streaming, Big data analysis and Visualization.

**Prerequisites:**

Programming skill and knowledge of object oriented programming, Data base management system and Exploratory data analytics.

**Course Outcomes (COs):**

At the end of the course the student should be able to:

1. Identify the key issues in big data management and applications in business and scientific computing.
2. Analyse the efficiency of big data storage solutions and integration strategies.
3. Apply stream data processing tools and techniques for real time data.
4. Develop big data processing models for a real-time application.
5. Apply data visualization techniques to interpret results.

**Course Articulation Matrix: Mapping of Course Outcomes (COs) with Program Outcomes (POs)**

|                                  |               |
|----------------------------------|---------------|
| Course Title: Big Data Analytics | Semester: 7   |
| Course Code: 26ECSC404           | Year: 2026-27 |

| Course Outcomes (COs) / Program Outcomes (POs)                                                           | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 |
|----------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|----|----|----|----|
| 1. Identify the key issues in big data management and applications in business and scientific computing. | H | M |   |   |   |   |   |   |   |    |    |    |    |
| 2. Analyse the efficiency of big data storage solutions and integration strategies.                      | H | M |   |   |   |   |   |   |   |    |    |    |    |
| 3. Apply stream data processing tools and techniques for real time data.                                 |   | H |   |   | H |   |   |   |   |    |    |    |    |
| 4. Develop big data processing models for a real-time application.                                       | H |   | H |   | H |   |   |   | M |    |    |    |    |
| 5. Apply data visualization techniques to interpret results.                                             |   | M | M |   | M |   |   |   |   |    |    |    |    |

Degree of compliance **L**: Low **M**: Medium **H**: High

**Competency addressed in the Course and corresponding Performance Indicators**

| Competency                                                                                                                                                                                                      | Performance Indicators                                                                                                                                         | Course Specific PI                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| 1.4:Demonstrate the competency in Computer Science and Engineering                                                                                                                                              | PI-1.4.2:Apply data engineering to model relationships, manage databases, ensure data integrity, and optimize solutions for actionable analytics and insights. | 1.4.2.1: Apply NoSQL schema with sharding/replication. Validate using MongoDB CRUD for storage.                             |
| 2.2:Demonstrate an ability to apply principles of mathematics, natural sciences, and engineering to formulate and analyse complex problems, incorporating literature review to support sustainable development. | PI-2.2.1:Decompose complex problems into interconnected sub-problems.                                                                                          | 2.2.1.1: Decompose cluster performance by examining replication and block size effects on scalability; document throughput. |
|                                                                                                                                                                                                                 | PI-2.2.4:Apply appropriate analytical,computational, experimental tools to evaluate problem scenarios with a focus on sustainable development.                 | 2.2.4.1:Apply HDFS+Hive RCFile and NoSQL through tests for optimal choice.                                                  |
| 3.2:Demonstrate an ability to generate a diverse set of alternative design solutions.                                                                                                                           | PI-3.2.2: Identify suitable criteria for evaluation of alternate design solutions.                                                                             | 3.2.2.1:Build a pipeline with sources, sinks, windows and transformations, then test it.                                    |
| 3.3:Demonstrate an ability to select an optimal design solution considering cost-effectiveness and environmental impact.                                                                                        | PI-3.3.2:Build models/prototypes to develop chosen design solutions.                                                                                           | 3.3.2.1:Develop tumbling and sliding windowing designs' accuracy-latency trade-offs.                                        |
| 5.2: Demonstrate an ability to apply discipline-specific tools, techniques, and resources.                                                                                                                      | PI-5.2.1:Create/modify parts and models using modern engineering and IT tools for the solution of complex engineering problems.                                | 5.2.1.1:Apply visualizations of trends from Hive, Pig, or Spark outputs that show query semantics.                          |
| 9.1:Demonstrate ability to comprehend technical literature, document project work, and communicate effectively.                                                                                                 | PI- 9.1.2:Organize and produce well-structured content in a document or presentation with a clear and logical flow.                                            | 9.1.2.1:Organise data preparation, models, metrics and visualizations supporting conclusions                                |

Eg: 1.2.3: Represents Program Outcome '1', Competency '2', and Performance Indicators '3'.

### Course Content

|                        |                                  |                        |
|------------------------|----------------------------------|------------------------|
| Course Code: 26ECSC404 | Course Title: Big Data Analytics |                        |
| L-T-P: 2-0-1           | Credits:3                        | Contact Hrs: 4hrs/week |
| ISA Marks: 50          | ESA Marks: 50                    | Total Marks:100        |
| Teaching Hrs: 30       | Practical Hrs:28                 | Exam Duration: 2 hrs   |

| Content                                                                                                                                                                                                                                                                                                                                                                    | Hrs    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <b>Unit - 1</b>                                                                                                                                                                                                                                                                                                                                                            |        |
| <b>1. Introduction:</b> Overview of Big data, Big Data Characteristics, Different Types of Data. Data Analytics, Data Analytics Life Cycle                                                                                                                                                                                                                                 | 05hrs  |
| <b>2. Big Data Storage:</b> Clusters, File Systems and Distributed File Systems, NoSQL, <b>No SQL Database:</b> Document-oriented, Column-oriented, Graph-based, MongoDB. Sharding, Replication, Combining Sharding and Replication. On Disk Storage Devices, In-memory Storage Devices.                                                                                   | 05 hrs |
| <b>3. Big Data Processing:</b> Parallel Data Processing, Distributed Data Processing, Hadoop, Map Reduce, Examples on MapReduce, Spark.                                                                                                                                                                                                                                    | 05 hrs |
| <b>Unit - 2</b>                                                                                                                                                                                                                                                                                                                                                            |        |
| <b>4. Stream Processing:</b> Introduction to Stream Processing-Batch Versus Stream Processing; Examples of Stream Processing; Scaling Up Data Processing; Distributed Stream Processing; Stream-Processing Model- Sources and Sinks, Immutable Streams Defined from One Another, Transformations and Aggregations, Window Aggregations, Stateless and Stateful Processing. | 05hrs  |
| <b>5. Big Data Analysis:</b> Pig- Introduction, Pig Primitive Data Types - Running Pig - Execution Modes of Pig – HDFS Commands - Relational Operators - Eval Function - Complex Data Types - Piggy Bank - User-Defined Functions - Parameter Substitution - Diagnostic Operator - Word Count Example using Pig - Pig at Yahoo! - Pig Versus Hive.                         | 05hrs  |
| <b>6. Big Data Visualization:</b> Hive – Introduction, Hive Architecture, Hive Data Types, Hive File Format, Hive Query Language (HQL), RCFile Implementation, User-Defined Function (UDF). Serialization and Deserialization.                                                                                                                                             | 05hrs  |

### Text Books

1. Thomas Erl, WajidKhattak, and Paul Buhler, Big Data Fundamentals Concepts, Drivers & Techniques, Prentice Hall, 2015.
2. SeemaAcharya, SubhashiniChellappan, Big Data and Analytics, Second Edition,Wiley India Pvt Ltd 2022
3. Gerard Maas and François Garillot, Stream Processing with Apache Spark Mastering Structured Streaming and Spark Streaming, O'REILLY,2019.

### References

1. Umit Demirbaga, Gagangeet Singh Aujla, Anish Jindal, Oğuzhan Kalyon, Big Data Analytics, Theory, Techniques, Platforms, and applications, Springer Nature Switzerland AG 2024. (eBook) <https://doi.org/10.1007/978-3-031-55639-5>.
2. Frank J Ohlhorst, Big Data and Analytics: Turning Big Data into Big Money, Wiley and SAS Business Series, 2012.
3. Colleen Mccue, Data Mining and Predictive Analysis: Intelligence Gathering and Crime Analysis, Elsevier, 2007.

### Evaluation Scheme In-Semester Assessment Scheme

| Assessment                       | Conducted for marks | Weightage in Marks |
|----------------------------------|---------------------|--------------------|
| In Semester Assessment-1 (ISA 1) | 30                  | 33                 |
| In Semester Assessment-2 (ISA 2) | 30                  |                    |
| Lab Assessment                   | 40                  | 17                 |
| <b>Total</b>                     |                     | <b>50</b>          |

### End-Semester Assessment Scheme

| Assessment   | Conducted for marks | Weightage in Marks |
|--------------|---------------------|--------------------|
| ESA (Theory) | 60                  | 33                 |
| ESA (Lab)    | 17                  | 17                 |
| <b>Total</b> |                     | <b>50</b>          |

Course Unitization for Minor Exams and End Semester Assessment

| Topics / Chapters         | Teaching Credits | No. of Questions in ISA-1 | No. of Questions in ISA-2 | No. of Questions in Lab Activity | No. of Questions in Theory ESA |
|---------------------------|------------------|---------------------------|---------------------------|----------------------------------|--------------------------------|
| <b>Unit I</b>             |                  |                           |                           |                                  |                                |
| 1. Introduction           | 5                | 1                         | --                        | --                               | 1                              |
| 2. Big Data Storage       | 5                | 1                         | --                        | 1                                | 1                              |
| 3. Big Data Processing    | 5                | 1                         | --                        | 1                                | 1                              |
| <b>Unit II</b>            |                  |                           |                           |                                  |                                |
| 4. Stream Processing      | 5                | --                        | 1                         | 1                                | 1                              |
| 5. Big Data Analysis      | 5                | --                        | 1                         | 1                                | 1                              |
| 6. Big Data Visualization | 5                | ---                       | 1                         | 1                                | 1                              |

**Note**

- Each Question carries 15 marks and might consist of sub-questions.
- Mixing of sub-questions from different chapters within a unit (Unit I and Unit II) is allowed in ISA I, ISAII, and ESA.
- Answer 4 full questions of 15 marks each (two full questions from Unit I, and two full questions from Unit II) out of 6 questions in ESA.

**Date:**

**HoD, CSE**

**Course Assessment Plan**

| Course Title: Big Data Analytics                                                                         |                         | Code: 26ECSC404    |        |                |              |                 |  |
|----------------------------------------------------------------------------------------------------------|-------------------------|--------------------|--------|----------------|--------------|-----------------|--|
| Course outcomes (COs)                                                                                    | Weightage in assessment | Assessment Methods |        |                |              | ESA-Theory Exam |  |
|                                                                                                          |                         | ISA I              | ISA II | Lab Assessment | ESA-Lab exam |                 |  |
| 1. Identify the key issues in big data management and applications in business and scientific computing. | 10%                     | ✓                  |        | ✓              | ✓            | ✓               |  |
| 2. Analyse the efficiency of big data storage solutions and integration strategies.                      | 25%                     | ✓                  |        |                | ✓            | ✓               |  |
| 3. Apply stream data processing tools and techniques for real time data.                                 | 25%                     | ✓                  | ✓      | ✓              | ✓            | ✓               |  |
| 4. Develop big data processing models for a real-time application.                                       | 20%                     |                    | ✓      | ✓              | ✓            | ✓               |  |
| 5. Apply data visualization techniques to interpret results.                                             | 20%                     |                    | ✓      | ✓              | ✓            | ✓               |  |
| Weightage                                                                                                | 100%                    | 37%                |        | 13%            | 50%          |                 |  |

## Course Activity and Rubrics

### Lab Activity plan

| Big Data Analytics Lab                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                    |     |              |                                          |         |     |       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|-----|--------------|------------------------------------------|---------|-----|-------|
| <p><b>Preamble:</b></p> <p>Data is created constantly, and at an ever-increasing rate. Mobile phones, social media, imaging technologies to determine a medical diagnosis—all these and more create new data, and that must be stored somewhere for some purpose. Devices and sensors automatically generate diagnostic information that needs to be stored and processed in real-time. Merely keeping up with this huge influx of data is difficult, but substantially more challenging is analyzing vast amounts of it, especially when it does not conform to traditional notions of data structure, to identify meaningful patterns and extract useful information. These challenges of the data deluge present the opportunity to transform business, government, science, and everyday life.</p> <p><b>Objective:</b> The student should be able to use Big Data and Analytics Frameworks and tools for handling, processing, and analyzing huge datasets.</p> <p><b>Team size:</b> 3- 4 (Only for Hadoop Implementation) for other activities individual assessments.</p> <p><b>Type:</b> Each batch will work for one distinct application area</p> |                                                                    |     |              |                                          |         |     |       |
| Sl. No.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Experiments                                                        | CO  | Blooms level | Timeline wrt COE                         | PI code | Hrs | Marks |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Hadoop Installation                                                | CO1 | L3           | 1 <sup>st</sup> &2 <sup>nd</sup> week    | 5.2.1   | 4   | Nil   |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Implementation of Replication and Sharding                         | CO2 | L3           | 3 <sup>rd</sup> & 4 <sup>th</sup> Week   | 3.3.2   | 4   | Nil   |
| 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Implementation of real time application using Hadoop.              | CO1 | L3           | 5 <sup>th</sup> to 8 <sup>th</sup> Week  | 3.3.2   | 8   | 20    |
| 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | MongoDB query practice                                             | CO2 | L3           | 9 <sup>th</sup> Week                     | 5.2.1   | 2   | Nil   |
| 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | MongoDB query evaluation for the given scenario and Certification. | CO2 | L3           | 10 <sup>th</sup> Week                    | 2.2.4   | 2   | 10    |
| 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Hive query practice                                                | CO5 | L3           | 11 <sup>th</sup> &12 <sup>th</sup> Week  | 5.2.1   | 4   | Nil   |
| 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Hive Query evaluation for the given scenario                       | CO2 | L3           | 13 <sup>th</sup> & 14 <sup>th</sup> Week | 5.2.1   | 4   | 10    |
| Total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                    |     |              |                                          |         | 28  | 40    |

### Assessment parameters and Rubrics for Lab Activity

| Phases                                                           | Excellent                                                                                                                                                                                                                                                                           | Good                                                                                                                                                                                                                            | Satisfactory                                                                                                                                                                                                 | Needs Improvement                                                                                                                                                                                         |
|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Implementation of real time application using Hadoop. (PI-3.3.2) | Demonstrates thorough understanding of the real-time problem, correctly configures the Hadoop environment, develops an efficient MapReduce application, produces accurate outputs, validates results, and confidently explains the implementation during the demonstration. (17-20) | Demonstrates good understanding of the problem, successfully configures Hadoop with minor issues, implements the application with few errors, generates mostly correct outputs, and provides satisfactory explanations. (13-16) | Demonstrates basic understanding of Hadoop concepts, requires assistance in implementation, partially executes the application, produces partially correct results, and provides limited explanation. (9-12) | Demonstrates inadequate understanding of the problem, is unable to configure Hadoop or implement the application correctly, produces incorrect results, and is unable to explain the implementation (0-8) |
| MongoDB query evaluation for the given scenario (PI-2.2.4)       | Correctly analyzes the given scenario, formulates accurate CRUD and aggregation queries, executes them successfully, validates the results, and clearly explains the solution. (9-10)                                                                                               | Correctly develops most queries with minor syntax or logical errors, executes them successfully, and provides a satisfactory explanation of the results. (7-8)                                                                  | Demonstrates basic understanding of MongoDB queries, develops simple queries with partial correctness, and provides limited interpretation of the results. (5-6)                                             | Unable to formulate or execute appropriate MongoDB queries, produces incorrect outputs, and cannot explain the query logic. (0-4)                                                                         |

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|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Hive Query (PI-5.2.1) | Correctly interprets the problem, develops efficient HiveQL queries, executes them successfully, validates outputs accurately, and confidently explains the query execution and results. (9-10) | Develops HiveQL queries with minor syntax or logical errors, executes them successfully, obtains mostly correct results, and provides satisfactory explanations. (7-8) | Demonstrates basic understanding of HiveQL, develops simple queries with partial correctness, and provides limited analysis of the results. (5-6) | Unable to develop or execute correct HiveQL queries, produces incorrect outputs, and cannot explain the solution or results. (0-4) |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|

#### Course Project Rubrics:

| Phases                                             | Exemplary (8-10)                                                                                                                                                | Satisfactory (5-7)                                                                                                                           | Needs Improvement (2-4)                                                                                                    | Not satisfactory (0-1)                                                                                                                           |
|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Problem Identification (PI - 2.2.1)                | The problem statement is clear, concise, and specific. It clearly identifies the problem, its context, and its scope.                                           | The problem statement is clear but comprises of ambiguities.                                                                                 | The problem is vaguely defined or lacks significant clarity, resulting in difficulties to understand the issue.            | The problem statement lacks clarity. It is unclear, doesn't provide context, and doesn't adequately convey the issue.                            |
| Data preparation (PI - 1.4.2)                      | Data sources are expertly-collected and highly relevant. Data is collected systematically and prepared with thorough cleaning, transformation, and integration. | The majority of the data sources are relevant and appropriate. But, minor issues exist with cleaning or transforming the data.               | Data is poorly prepared with major issues in cleaning, transformation, or handling missing data.                           | Sources of data are missing or irrelevant. inadequate data preparation, with significant problems in collecting data and cleansing.              |
| Model Selection (PI - 2.2.4)                       | Decision and design recommendations have a strong base and justification.                                                                                       | Decision and design recommendation are reasonable.                                                                                           | Design and model are inappropriate or poorly selected. The approach lacks structure.                                       | Decision and design recommendation are not relevant.                                                                                             |
| Implementation of real time application (PI-5.2.1) | Selection of appropriate analytical tools and techniques. Developed code that follows the design specification.                                                 | Selection of appropriate analytical tools and techniques. Developed code that follows the design specification, but can be further improved. | Significant technical problems or major functionality issues. The project may not work as intended or lacks core features. | Needs improvement in the analytical tools and/or software engineering techniques. The developed code is not mapping to the design specification. |

|                                    |                                                                                                                                   |                                                                                                                                   |                                                                                                     |                                                                                 |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Presentation and Report (PI-9.1.2) | The presentation is professionally delivered. The speaker exhibits strong confidence. Demo is delivered within the allotted time. | The presentation is understandable but may lack engagement. The speaker shows some confidence but may struggle with articulation. | There is ambiguity in the presentation. The speaker is unconfident. There is clear time management. | Communication is ineffective. Time management may be inconsistent.              |
|                                    | Documentation is comprehensive, well-organized.                                                                                   | Documentation is ordered and comprehensive. could be structured better.                                                           | Documentation is unclear, poorly structured, Key points don't carry clarity.                        | Documentation is desperately organized, with significant problems with clarity. |

### Chapter-wise Plan

|                                                             |                             |
|-------------------------------------------------------------|-----------------------------|
| Course Code and Title: <b>26ECSC404/ Big Data Analytics</b> |                             |
| Chapter Number and Title: <b>1. Introduction</b>            | Planned Hours: <b>05hrs</b> |

#### Learning Outcomes:-

At the end of the topic the student should be able to:

| Topic Learning Outcomes                                                            | COs | BL | CA Code |
|------------------------------------------------------------------------------------|-----|----|---------|
| 1.Differentiate between different contemporary practices in data analytics.        | CO1 | L3 | 1.4     |
| 2.Analyze the importance of big data and the challenges of Big Data and Analytics; | CO1 | L3 | 2.2     |
| 3.Explain the Importance of data analysis for business intelligence applications   | CO1 | L2 | 1.4     |
| 4.Explain the various characteristics of bigdata                                   | CO1 | L2 | 1.4     |

|                                                  |
|--------------------------------------------------|
| Lesson Schedule                                  |
| Class No. - Portion covered per hour             |
| 1.Overview of Big data, Big Data Characteristics |
| 2. Different Types of Data                       |
| 3. Data Analytics                                |
| 4. Data Analytics Life Cycle                     |
| 5. Data Analytics Life Cycle                     |

#### Review Questions

| Sl.No. - Questions                                                                                              | TLOs | BL | PI Code |
|-----------------------------------------------------------------------------------------------------------------|------|----|---------|
| 1. Differentiate between descriptive and prescriptive data analytics.                                           | TLO1 | L3 | 1.4.2   |
| 2. Explain how BI enables an organization to gain insight into the performance of an enterprise.                | TLO3 | L3 | 2.2.4   |
| 3. Explain the differences between BI and Data Science.                                                         | TLO3 | L2 | 1.4.2   |
| 4. Explain the emerging big data ecosystem.                                                                     | TLO4 | L2 | 1.4.2   |
| 5. What are the three characteristics of Big Data, and what are the main considerations in processing Big Data? | TLO4 | L2 | 1.4.2   |
| 6. Differentiate between the different types of data processed by big data solutions.                           | TLO2 | L3 | 2.2.4   |

|                                                             |                             |
|-------------------------------------------------------------|-----------------------------|
| Course Code and Title: <b>26ECSC404/ Big Data Analytics</b> |                             |
| Chapter Number and Title: <b>2. Big Data Storage</b>        | Planned Hours: <b>05hrs</b> |

**Learning Outcomes:-**

**At the end of the topic the student should be able to:**

| Topic Learning Outcomes                                                                                    | COs | BL | CA Code |
|------------------------------------------------------------------------------------------------------------|-----|----|---------|
| 1.Understand the storage of large files spread across the nodes of a cluster in a distributed file system; | CO2 | L2 | 1.4     |
| 2.Explain Document oriented,column oriented and Graph based databases.                                     | CO2 | L2 | 1.4     |
| 3.Write Mongo query language to perform CRUD operations on a given dataset in MongoDB.                     | CO2 | L3 | 1.4     |
| 4.Explain how sharding provides partial tolerance toward failures and horizontal scalability;              | CO2 | L3 | 2.2     |
| 5.Explain how Replication provides scalability, data availability and fault tolerance;                     | CO2 | L3 | 1.4     |

|                                                                               |
|-------------------------------------------------------------------------------|
| Lesson Schedule                                                               |
| Class No. - Portion covered per hour                                          |
| 1. Clusters; File Systems; Distributed File Systems                           |
| 2.No SQL Databases: Document-oriented, Column-oriented, Graph-based, MongoDB. |
| 3. Sharding                                                                   |
| 4. Replication; Combining Sharding and Replication                            |
| 5. On Disk Storage Devices, In-memory Storage Devices                         |

**Review Questions**

| Sl.No. - Questions                                                                                                        | TLOs | BL | PI Code |
|---------------------------------------------------------------------------------------------------------------------------|------|----|---------|
| 1. Explain how a large file is stored on a distributed file system.                                                       | TLO1 | L2 | 1.4.2   |
| 2.How sharding differs with replication? Illustrate with example.                                                         | TLO3 | L3 | 1.4.2   |
| 3. Explain how sharding helps to achieve horizontal scalability.                                                          | TLO3 | L2 | 1.4.2   |
| 4.Explain how data availability and fault tolerance achieved through replication in distributed data storage? Illustrate. | TLO4 | L3 | 2.2.4   |
| 5. Illustrate the use of NoSQL for ETL operations on the dataset stored on a DFS.                                         | TLO3 | L3 | 1.4.2   |
| 6.Write a MongoDB Query language to create "Library " collection and perform CURD operations for the Library system.      | TLO3 | L3 | 1.4.2   |

|                                                             |                             |
|-------------------------------------------------------------|-----------------------------|
| Course Code and Title: <b>26ECSC404/ Big Data Analytics</b> |                             |
| Chapter Number and Title: <b>3. Big Data Processing</b>     | Planned Hours: <b>05hrs</b> |

### Learning Outcomes:-

At the end of the topic the student should be able to:

| Topic Learning Outcomes                                                                                                                     | COS | BL | CA Code |
|---------------------------------------------------------------------------------------------------------------------------------------------|-----|----|---------|
| 1. Explain the principles of parallel data processing and their role in improving the scalability and performance of Big Data applications. | CO3 | L2 | 2.2     |
| 2. Describe distributed data processing.                                                                                                    | CO3 | L2 | 2.2     |
| 3. Illustrate the processing and storage capability of Hadoop framework                                                                     | CO3 | L3 | 2.2     |
| 4. Analyze the importance Hadoop Distributed File System (HDFS)                                                                             | CO3 | L3 | 2.2     |
| 5. Describe the processing of real time data using Spark                                                                                    | CO3 | L2 | 2.2     |

|                                      |
|--------------------------------------|
| Lesson Schedule                      |
| Class No. - Portion covered per hour |
| 1. Parallel Data Processing          |
| 2. Distributed Data Processing       |
| 3. Hadoop ,Map Reduce                |
| 4. Examples on MapReduce             |
| 5. Spark                             |

### Review Questions

| Sl.No. - Questions                                                                                   | TLOs | BL | PI Code |
|------------------------------------------------------------------------------------------------------|------|----|---------|
| 1. Illustrate the process of parallel data processing with example.                                  | TLO1 | L1 | 2.2.1   |
| 2. Analyze the processing of data with distributed approach with example.                            | TLO2 | L2 | 2.2.1   |
| 3. Draw and explain the framework of Hadoop.                                                         | TLO3 | L3 | 2.2.1   |
| 4. Write mapper and reducer function in Java to count the number of words in the chosen input file.  | TLO4 | L3 | 2.2.4   |
| 5. How the processing of continuous data is carried out using Spark. Discuss the suitable technique. | TLO5 | L3 | 2.2.4   |

|                                                             |                             |
|-------------------------------------------------------------|-----------------------------|
| Course Code and Title: <b>26ECSC404/ Big Data Analytics</b> |                             |
| Chapter Number and Title: <b>4. Stream Processing</b>       | Planned Hours: <b>05hrs</b> |

**Learning Outcomes:-**

**At the end of the topic the student should be able to:**

| Topic Learning Outcomes                                                 | COs | BL | CA Code |
|-------------------------------------------------------------------------|-----|----|---------|
| 1.Understand how stream processing is different from batch processing;  | CO3 | L2 | 2.2     |
| 2.Apply stream processing tools and techniques for big data processing; | CO3 | L3 | 2.2     |
| 3.Discuss immutable stream processing;                                  | CO3 | L2 | 2.2     |
| 4.Elucidate transformations and aggregations performed on stream data;  | CO3 | L2 | 2.2     |

|                                                                                  |
|----------------------------------------------------------------------------------|
| Lesson Schedule                                                                  |
| Class No. - Portion covered per hour                                             |
| 1. Introduction to Stream Processing, Batch Versus Stream Processing             |
| 2. Examples of Stream Processing, Scaling Up Data Processing                     |
| 3.Distributed Stream Processing; .Stream-Processing Model- Sources and Sinks     |
| 4. Immutable Streams Defined from One Another, Transformations and Aggregations, |
| 5. Window Aggregations, Stateless and Stateful Processing.                       |

**Review Questions**

| Sl.No. - Questions                                                                                                          | TLOs | BL | PI Code |
|-----------------------------------------------------------------------------------------------------------------------------|------|----|---------|
| 1. How stream processing is different from batch processing.                                                                | TLO1 | L2 | 2.2.1   |
| 2. Illustrate how distributed processing of stream data helps scale up of data processing facilitating big data processing. | TLO2 | L3 | 2.2.1   |
| 3. Interpret how the tumbling windows are used for window aggregation.                                                      | TLO3 | L2 | 2.2.4   |
| 4. Differentiate between Stateless and Stateful Processing                                                                  | TLO4 | L3 | 2.2.1   |

|                                                             |                             |
|-------------------------------------------------------------|-----------------------------|
| Course Code and Title: <b>26ECSC404/ Big Data Analytics</b> |                             |
| Chapter Number and Title: <b>5. Big Data Analysis</b>       | Planned Hours: <b>05hrs</b> |

**Learning Outcomes:-**

**At the end of the topic the student should be able to:**

| Topic Learning Outcomes                 | COs | BL | CA Code |
|-----------------------------------------|-----|----|---------|
| 1.Discuss the data types of Pig         | CO5 | L2 | 2.2     |
| 2.Illustrate HDFS Commands              | CO5 | L3 | 3.2     |
| 3.Differentiate between Pig versus Hive | CO5 | L3 | 2.2     |

|                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------|
| Lesson Schedule                                                                                                            |
| Class No. - Portion covered per hour                                                                                       |
| 1.Pig- Introduction, Pig Primitive Data Types - Running Pig, Execution Modes of Pig – HDFS Commands - Relational Operators |
| 2. Eval Function - Complex Data Types                                                                                      |
| 3. Piggy Bank - User-Defined Functions                                                                                     |
| 4. Parameter Substitution - Diagnostic Operator, Word Count Example using Pig                                              |
| 5.Pig at Yahoo! - Pig Versus Hive                                                                                          |

**Review Questions**

| Sl.No. - Questions                                                  | TLOs | BL | PI Code |
|---------------------------------------------------------------------|------|----|---------|
| 1.Explain the data types of Pig.                                    | TLO1 | L2 | 2.2.1   |
| 2.Write the functions of Piggy bank.                                | TLO3 | L2 | 2.2.1   |
| 3.Illustrate HDFS commands with example.                            | TLO2 | L3 | 3.3.2   |
| 4.Write a program to count the number of words in Pig.              | TLO1 | L3 | 2.2.1   |
| 5.Discuss the relational operators with example.                    | TLO1 | L2 | 2.2.1   |
| 6.Why do we need MapReduce during Pig programming?                  | TLO2 | L3 | 3.3.2   |
| 7.Describe the architecture of Apache Pig                           | TLO3 | L2 | 2.2.1   |
| 8.Illustrate the complex data types in pig? Illustrate with example | TLO1 | L2 | 2.2.1   |
| 9.What are the different execution modes available in Pig ?         | TLO3 | L2 | 2.2.1   |

|                                                             |                             |
|-------------------------------------------------------------|-----------------------------|
| Course Code and Title: <b>26ECSC404/ Big Data Analytics</b> |                             |
| Chapter Number and Title: <b>6. Big Data Visualization</b>  | Planned Hours: <b>05hrs</b> |

### Learning Outcomes:-

At the end of the topic the student should be able to:

| Topic Learning Outcomes                                 | COs | BL | CA Code |
|---------------------------------------------------------|-----|----|---------|
| 1.Explain the need for Hive in Big Data Analytics.      | CO5 | L2 | 3.2     |
| 2.Describe the Architecture of HIVE and Data Types      | CO5 | L2 | 2.2     |
| 3.Understand the Importance of Hive file formats.       | CO5 | L2 | 2.2     |
| 4.Write HIVE Query Language to query a dataset in HIVE. | CO5 | L3 | 3.2     |

|                                                                    |
|--------------------------------------------------------------------|
| Lesson Schedule                                                    |
| Class No. - Portion covered per hour                               |
| 1.Hive – Introduction, Hive Architecture                           |
| 2. Hive Data Types, Hive File Format                               |
| 3. Hive Query Language (HQL)                                       |
| 4. RCFile Implementation                                           |
| 5. User-Defined Function (UDF). Serialization and Deserialization. |

### Review Questions

| Sl. No. - Questions                                                                                                                                                                                                                                                                                                         | TLOs | BL | PI Code |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----|---------|
| 1. Discuss the features of HIVE.                                                                                                                                                                                                                                                                                            | TLO1 | L1 | 3.2.2   |
| 2. Explain with example HIVE is a data warehousing tool.                                                                                                                                                                                                                                                                    | TLO1 | L2 | 3.2.2   |
| 3. Explain with neat diagram data units arranged in Hive.                                                                                                                                                                                                                                                                   | TLO2 | L2 | 2.2.4   |
| 4. With diagram describe HIVE architecture                                                                                                                                                                                                                                                                                  | TLO3 | L2 | 2.2.4   |
| 5. Explain with example different file formats supported by HIVE.                                                                                                                                                                                                                                                           | TLO3 | L2 | 2.2.4   |
| 6. Explain DDL and DML in HIVE.                                                                                                                                                                                                                                                                                             | TLO2 | L2 | 2.2.4   |
| 7. Write HiveQL statement to create a data file for below schemas:<br>Order: CustomerId, ItemID, ItemName, OrderDate, DeliveryDate<br>Customer: CustomerId, CustomerName, Address, City, State, Country<br>Create a table for Order and Customer data.<br>Write HiveQL to find the number of items bought by each customer. | TLO4 | L2 | 3.3.2   |

| Model Question Paper for Minor Examination (ISA-I)                           |                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                  |     |    |         |
|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-----|----|---------|
| Course Code: 26ECSC404                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                        | Course Title: Big Data Analytics |     |    |         |
| Duration (H:M): 1 hr                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                        | Max. Marks: 30                   |     |    |         |
| Note: Answer any two full questions. Each full question carries equal marks. |                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                  |     |    |         |
|                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                  |     |    |         |
| Q.No                                                                         | Questions                                                                                                                                                                                                                                                                                                                                                                                                                              | Marks                            | CO  | BL | PI Code |
| Q.No                                                                         | Questions                                                                                                                                                                                                                                                                                                                                                                                                                              | Marks                            | CO  | BL | PI Code |
| 1a                                                                           | A smart city collects large amounts of data from traffic cameras, GPS-enabled vehicles, parking sensors, weather stations, public transport, and citizen applications. The city administration wants to analyze this data to reduce traffic congestion and improve public services. Analyze how the 5Vs of Big Data affect the selection of a suitable Big Data solution for this scenario. Justify your answer.                       | 8                                | CO1 | L3 | 1.4.2   |
| 1b                                                                           | A bank collects large volumes of customer transaction records, account activities, and payment patterns. The bank aims to identify unusual transactions and minimize financial losses caused by fraudulent activities.Analyze the given scenario and recommend suitable Big Data analytics techniques for detecting fraudulent transactions. Justify the selection of each technique based on its applicability to the banking domain. | 7                                | CO1 | L3 | 1.4.2   |
| 2a                                                                           | A university implements an HDFS cluster to manage student records, examination results, and digital learning resources. Analyze the effect of different HDFS block sizes and replication factors on fault tolerance, storage utilization, data access, and overall processing performance.                                                                                                                                             | 8                                | CO2 | L3 | 2.2.1   |
| 2b                                                                           | Formulate a MongoDB schema with sharding and replication for the given Student database. Validate the solution using CRUD operations.                                                                                                                                                                                                                                                                                                  | 7                                | CO2 | L3 | 2.2.4   |
| 3a                                                                           | Assess Hadoop components for a large-scale data processing scenario and recommend an appropriate processing strategy.                                                                                                                                                                                                                                                                                                                  | 8                                | CO3 | L3 | 2.2.4   |
| 3b                                                                           | Write Mapper and Reducer logic for counting occurrences of each product sold in an online shopping dataset.                                                                                                                                                                                                                                                                                                                            | 7                                | CO3 | L3 | 2.2.4   |

| Model Question Paper for Minor Examination (ISA-II)                          |                                                                                                                                                                                                                                                                                                                                                                                                                       |                                  |     |    |         |
|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-----|----|---------|
| Course Code: 26ECSC404                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                       | Course Title: Big Data Analytics |     |    |         |
| Duration (H:M): 1 hr                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                       | Max. Marks: 30                   |     |    |         |
| Note: Answer any two full questions. Each full question carries equal marks. |                                                                                                                                                                                                                                                                                                                                                                                                                       |                                  |     |    |         |
|                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                       |                                  |     |    |         |
| Q.No                                                                         | Questions                                                                                                                                                                                                                                                                                                                                                                                                             | Marks                            | CO  | BL | PI Code |
| Q.No                                                                         | Questions                                                                                                                                                                                                                                                                                                                                                                                                             | Marks                            | CO  | BL | PI Code |
| 1a                                                                           | An e-commerce platform wants to personalize product recommendations in real time while also analyzing monthly sales trends for inventory planning.Discuss how batch processing and stream processing can be applied in this retail scenario. Which model should be prioritized for customer engagement, and why?                                                                                                      | 8                                | CO3 | L3 | 2.2.4   |
| 1b                                                                           | Develop a stream-processing pipeline using sources, sinks, transformations and window operations for the given IoT scenario.                                                                                                                                                                                                                                                                                          | 7                                | CO3 | L3 | 3.3.2   |
| 2a                                                                           | Construct a Pig Latin program to compute product-wise sales from the given dataset and validate the output.                                                                                                                                                                                                                                                                                                           | 8                                | CO5 | L3 | 3.3.2   |
| 2b                                                                           | A social media company collects raw clickstream logs and user activity data. It needs to preprocess this unstructured data for sentiment analysis, while also generating structured reports on trending topics.Assess the suitability of Apache Pig and Apache Hive for this social media analytics scenario. Which framework is more appropriate for preprocessing raw logs, and why?                                | 7                                | CO5 | L3 | 3.3.2   |
| 3a                                                                           | Evaluate Hive file formats for the given data warehouse and recommend the optimal storage format.                                                                                                                                                                                                                                                                                                                     | 8                                | CO5 | L3 | 2.2.4   |
| 3b                                                                           | An online learning platform stores data in Student(StudentID, Name, CourseID) and Course(CourseID, CourseName) tables. The platform wants to know how many students are enrolled in each course.<br>i)Write HiveQL statements to create the Student and Course tables<br>ii)Display the course-wise student count.<br>iii) Display courses with no students enrolled.<br>iv) Display the top 3 courses by enrollment. | 7                                | CO5 | L3 | 2.2.4   |

| Model Question Paper for End Semester Examination (ESA)                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |                                  |    |         |
|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------------------------------|----|---------|
| Course Code: 26ECSC404                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                         |       | Course Title: Big Data Analytics |    |         |
| Duration: 2 hrs                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                         |       | Max. Marks: 60 marks             |    |         |
| <b>Note: Answer any two full questions from each unit-I and unit-II and one full question from unit-III</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |                                  |    |         |
| UNIT 1                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |                                  |    |         |
| Q.No                                                                                                        | Questions                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks | CO                               | BL | PI Code |
| Q.No                                                                                                        | Questions                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks | CO                               | BL | PI Code |
| 1a                                                                                                          | A smart city collects large amounts of data from traffic cameras, GPS-enabled vehicles, parking sensors, weather stations, public transport, and citizen applications. The city administration wants to analyze this data to reduce traffic congestion and improve public services. Analyze how the 5Vs of Big Data affect the selection of a suitable Big Data solution for this scenario. Justify your answer.                        | 8     | CO1                              | L3 | 1.4.2   |
| 1b                                                                                                          | A bank collects large volumes of customer transaction records, account activities, and payment patterns. The bank aims to identify unusual transactions and minimize financial losses caused by fraudulent activities. Analyze the given scenario and recommend suitable Big Data analytics techniques for detecting fraudulent transactions. Justify the selection of each technique based on its applicability to the banking domain. | 7     | CO1                              | L3 | 1.4.2   |
| 2a                                                                                                          | A university implements an HDFS cluster to manage student records, examination results, and digital learning resources. Analyze the effect of different HDFS block sizes and replication factors on fault tolerance, storage utilization, data access, and overall processing performance.                                                                                                                                              | 8     | CO2                              | L3 | 2.2.4   |
| 2b                                                                                                          | Formulate a MongoDB schema with sharding and replication for the given Student database. Validate the solution using CRUD operations.                                                                                                                                                                                                                                                                                                   | 7     | CO2                              | L3 | 2.2.4   |
| 3a                                                                                                          | Assess Hadoop components for a large-scale data processing scenario and recommend an appropriate processing strategy.                                                                                                                                                                                                                                                                                                                   | 8     | CO3                              | L3 | 2.2.4   |
| 3b                                                                                                          | Write Mapper and Reducer logic for counting occurrences of each product sold in an online shopping dataset.                                                                                                                                                                                                                                                                                                                             | 7     | CO5                              | L3 | 2.2.4   |
| UNIT 2                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |                                  |    |         |
| Q.No                                                                                                        | Questions                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks | CO                               | BL | PI Code |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                         |   |     |    |       |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----|----|-------|
| 4a | An e-commerce platform wants to personalize product recommendations in real time while also analyzing monthly sales trends for inventory planning. Discuss how batch processing and stream processing can be applied in this retail scenario. Which model should be prioritized for customer engagement, and why?                                                                                                       | 8 | CO3 | L3 | 2.2.4 |
| 4b | Develop a stream-processing pipeline using sources, sinks, transformations and window operations for the given IoT scenario.                                                                                                                                                                                                                                                                                            | 7 | CO3 | L3 | 3.3.2 |
| 5a | Construct a Pig Latin program to compute product-wise sales from the given dataset and validate the output.                                                                                                                                                                                                                                                                                                             | 8 | CO5 | L3 | 3.3.2 |
| 5b | A social media company collects raw clickstream logs and user activity data. It needs to preprocess this unstructured data for sentiment analysis, while also generating structured reports on trending topics. Assess the suitability of Apache Pig and Apache Hive for this social media analytics scenario. Which framework is more appropriate for preprocessing raw logs, and why?                                 | 7 | CO5 | L3 | 3.3.2 |
| 6a | Evaluate Hive file formats for the given data warehouse and recommend the optimal storage format.                                                                                                                                                                                                                                                                                                                       | 8 | CO5 | L3 | 2.2.4 |
| 6b | An online learning platform stores data in Student(StudentID, Name, CourseID) and Course(CourseID, CourseName) tables. The platform wants to know how many students are enrolled in each course.<br>i) Write HiveQL statements to create the Student and Course tables<br>ii) Display the course-wise student count.<br>iii) Display courses with no students enrolled.<br>iv) Display the top 3 courses by enrollment. | 7 | CO5 | L3 | 2.2.4 |